

Congruence Notes for Math 135

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Contents

1	Basics	2
1.1	Even and odd integers	2
1.2	Integers mod 5	3
1.3	Fields	4
1.4	Fields	4
1.5	Integers mod 6	4
1.6	Inverses (mod p)	5
2	Linear Congruences	5
2.1	$ax \equiv b \pmod{p}$	5
2.2	$ax \equiv b \pmod{n}$	6
3	Quadratic Congruences	6
3.1	Square roots	6
3.2	Quadratics mod p	7
3.3	Quadratics mod n	8
4	Divisibility	9
5	Inverses (mod p)	10
5.1	(gcd)	10
5.2	Finding gcd by factorisation	11
5.3	The "division algorithm"	11
5.4	Euclid’s algorithm	11
5.5	Pf. of Euclid’s Alg.	12
5.6	gcd and $a^{-1} \pmod{p}$	13
6	Exercises	14
7	Answers	15

1 Basics

1.1 Even and odd integers

The sum of two even integers is even, the sum of two odd integers is even, and the sum of an even and an odd integer is odd. Similarly, the product of two even integers is even, the product of two odd integers is odd, and the product of an even and an odd integer is even. This information can be summarized in the addition and multiplication tables:

+	Even	Odd
Even	Even	Odd
Odd	Odd	Even

$\times$	Even	Odd
Even	Even	Even
Odd	Even	Odd

Table 1: Multiplication and Addition for odd and even integers.

We have divided the natural numbers into two sets:

- the even integers 0, 2, 4, 6, ...
- the odd integers 1, 3, 5, 7, ...

and observed that any two numbers chosen from the same set behave alike with regard to whether the answer to an addition or multiplication is even or odd.

We call these sets "**residue classes modulo 2**". The set of even integers 0, 2, 4, 6, ... is the set of integers which leave a remainder (or "residue") of 0 when divided by 2. We say that a number in this set "is congruent to 0 modulo 2", written, e.g., as $6 \equiv 0 \pmod{2}$. Similarly, any number chosen from the set of odd integers 1, 3, 5, 7, ... leaves a residue (remainder) of 1 when divided by 2; we say it is congruent to 1 modulo 2, written obviously as, e.g., $5 \equiv 1 \pmod{2}$.

Since all the numbers in a given one of these sets behave the same way with respect to addition and multiplication, we can choose a typical representative of each set to illustrate the behavior of any number in that set. If we choose 0 to be the typical element from the set of even integers, and 1 to be the typical element of the set of odd integers, we can write our addition and multiplication tables in terms of 0 and 1.

+	0	1
0	0	1
1	1	0

$\times$	0	1
0	0	0
1	0	1

Table 2: Multiplication and addition modulo 2.

1.2 Integers mod 5

Suppose we are interested not just in whether an answer is even or odd, but in the remainder (residue) after division by 5. Now the possible remainders (residues) mod 5 are 0, 1, 2, 3, 4, yielding the residue classes

- 0, 5, 10, 15, ...
- 1, 6, 11, 16, ...
- 2, 7, 12, 17, ...
- 3, 8, 13, 18, ...
- 4, 9, 14, 19, ...

with 0, 1, 2, 3, 4 as typical representatives mod 5.

Example: Construct the addition and multiplication tables mod 5.

Solution:

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

×	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

Table 3: Multiplication and addition modulo 5.

These tables of course only summarise results which we can easily work out whenever we need them. With these tables or otherwise we can do many of the arithmetic calculations that we can do with real numbers.

Example: Find $4 + 3 \cdot 2 \pmod{5}$

Answer: From the table, $3 \cdot 2 = 1$, and $4 + 1 = 0$, so $4 + 3 \cdot 2 \equiv 0 \pmod{5}$.

Example: Find $2 - 4 \pmod{5}$

Solution: If $2 - 4 = x$, $2 = x + 4$. From the addition table, $3 + 4 \equiv 2 \pmod{5}$, so $x = 3$.

Example: Find $3^4 \pmod{5}$

Solution: $3^2 \equiv 4 \equiv -1 \pmod{5}$, so $3^4 \equiv (-1)^2 \equiv 1 \pmod{5}$ Less nicely, $3^2 \equiv 4 \pmod{5}$, so $3^4 = (3^2)^2 \equiv 4^2 \equiv 1 \pmod{5}$ (Notice that it was convenient to use $-1 \equiv 4 \pmod{5}$ in this case).

Example: Solve $2y \equiv 3 \pmod{5}$

Solution: From the table, $2 \cdot 4 \equiv 3 \pmod{5}$, so $y = 4$ or: Multiply both sides by 3: $3 \cdot 2 \cdot y \equiv 3 \cdot 3$, so $1 \cdot y \equiv 4$, or $y \equiv 4 \pmod{5}$.

1.3 The "field" of real numbers

The real numbers (technically, the real numbers together with the operations of addition and multiplication) are our most familiar example of an algebraic structure called a "field". The numbers (elements) in a field obey eleven axioms that we are familiar with from our operations with real numbers (e.g., $a + b = b + a$; $a \cdot (b + c) = a \cdot b + a \cdot c$). The complex numbers also form a field, which is why we introduced complex numbers by inventing a new number i "which obeys all of the usual rules of arithmetic".

One key property of fields will be specially important to us in working with congruences:

In a field (e.g., the real numbers), every non-zero number has a "*multiplicative inverse*", ie, given x , there is an x^{-1} such that $x \cdot x^{-1} = 1$. To divide by a number, we multiply by its inverse.

1.4 The integers mod 5 form a field

We won't prove this assertion we'll simply do as we did when we introduced complex numbers and manipulate them as we do real numbers, making whatever obvious changes are necessary.

Consider again the solution to our last example: "To solve $2y \equiv 3 \pmod{5}$, multiply both sides of the equation (more precisely, the congruence) by 3". We now see that since $2 \cdot 3 \equiv 1 \pmod{5}$, 3 is the inverse of 2 $\pmod{5}$, so multiplying by 3 is the equivalent of dividing by 2 in arithmetic mod 5.

So we are now in a position to solve any equation of the form $ax \equiv b \pmod{5}$, given a and b . Of course, this isn't a great achievement; there are only 5 possible values for a , and if $a = 0$ then (just as with real numbers) $0 \cdot x = 0$ for all x , so there are really only 4 interesting values for a .

1.5 Integers mod 6

Example: Find the inverse of 2 $\pmod{6}$.

Solution: Construct a multiplication table $\pmod{6}$. From it, we see that no multiple of 2 is congruent to 1 $\pmod{6}$. (Of course, even without the table this is obvious from the arithmetic of integers; 2 is even, so every multiple of 2 is even, but a number congruent to 1 $\pmod{6}$ must be odd!-why?).

Conclusion: The integers mod 6 do not form a field, and more generally:

Integers mod p form a field (p prime), integers mod n (n composite) do not.

Not only do the integers modulo 5 form a field, but the set of integers modulo p form a field for any prime p . We'll accept this without proof.

Notation

We often write $\mathbb{Z}_n$ as a convenient notation for "integers modulo n ".

Convention

As is usual in number theoretical work, we will assume that p represents a prime (unless we are explicitly told otherwise). Other symbols will represent numbers that may or may not be prime.

1.6 Finding inverses (mod p)

Method 1: Look at the multiplication table; e.g., $2 \cdot 3 \equiv 1 \pmod{5}$, so $2^{-1} = 3$ and $3^{-1} = 2$.

Method 2: Intelligent trial and error: Consider integers mod 7: $2 \cdot 4 \equiv 1 \pmod{7}$, so 2 and 4 are inverses of each other. Similarly $3 \cdot 5 \equiv 1 \pmod{7}$, so 3 and 5 are inverses of each other. What is the inverse of 6 (mod 7)?

Example: Find $4^{-1} \pmod{13}$

Solution: Since $4 \cdot 10 = 40 \equiv 1 \pmod{13}$, 10 and 4 are inverses of each other mod 13.

Exercise 1: Find 5 pairs of integers that are inverses of each other mod 13. Why can't you find a sixth pair?

(Answers to some of the exercises appear at the end of these notes. Before looking at the answers, you should solve the exercises as far as you can, i.e., find your solution and use any checking method you can think of to verify your solutions. After you have verified your solution, look at the answers I have provided, to see whether I have a different and/or nicer method of solution than you.)

In all cases (mod p), 0 has no inverse, and ± 1 are their own inverses (since $1^2 = 1$ and $(-1)^2 = 1$). All other numbers come in pairs which are inverses of each other. E.g., modulo 7, $2 \cdot 4 = 8 \equiv 1$, so 2 and 4 are inverses of each other; and $3 \cdot 5 = 15 \equiv 1$, so 3 and 5 are inverses of each other.

2 Linear Congruences

2.1 Solving $ax \equiv b \pmod{p}$

Example: Solve $3x \equiv 4 \pmod{7}$.

Solution: Obviously $3 \cdot 6 \equiv 4 \pmod{7}$, so $x = 6$.

Or: $3^{-1} \pmod{7} = 5$, $\implies 5 \cdot 3 \cdot x = 4 \cdot 5$, so $x \equiv 6 \pmod{7}$

Exercise 2: Solve $6x \equiv 4 \pmod{13}$

But suppose I had posed the problem: Solve $127x \equiv 157 \pmod{179}$. To construct a multiplication table $\pmod{179}$ would be absurd. But how else could you find the inverse of 127 $\pmod{179}$ at this stage? It happens that $127^{-1} \equiv 148 \pmod{179}$. So we clearly need a better way to find inverses than the trial and error methods we have so far. We'll come back to this question later.

2.2 Solving $ax \equiv b \pmod{n}$, n composite

Consider the multiplication table $\pmod{6}$. As we've seen before, the integers $\pmod{n}$ do not form a field for composite n . Here only 1 and 5 have an inverse (viz, themselves), so we can't guarantee that we can solve every equation of the form $ax \equiv b \pmod{n}$, and in general we can't. There are in fact techniques for "reducing" congruences modulo n , where n is composite, and we'll explore this avenue later. (We will see that $ax \equiv b \pmod{m}$ has solutions if and only if the greatest common divisor of a and m also divides b ; more capable students should try to prove this result now. We say more about greatest common divisors later.)

Example: Solve $4x \equiv 2 \pmod{6}$.

Solution: Since $4 \cdot 2 \equiv 2 \pmod{6}$ and $4 \cdot 5 \equiv 2 \pmod{6}$, the solutions to the equation are $x \equiv 2$ or 5 .

Note that the non-existence of $4^{-1} \pmod{6}$ doesn't mean that the congruence has no solution; it simply means that we can't solve the congruence by using 4^{-1} .

Example: Solve $4x \equiv 3 \pmod{6}$

Solution: From a multiplication table mod 6, there is no x for which $4x \equiv 3$. (This is obvious; in ordinary integers, every multiple of 4 is even, while every integer congruent to 3 mod 6 is odd.) So the congruence has no solution.

This raises an obvious question to a mathematician-can we characterise the cases where solutions exist and where they don't? The answer is "Yes", as we'll see.

3 Quadratic Congruences

3.1 Square roots mod n (n prime or composite)

If a is a square root of $x \pmod{n}$, then $a^2 \equiv x \pmod{n}$ by definition.

Theorem 3.1. If a is a square root of $x \pmod{n}$, so is $n - a$.

Proof. Modulo n , $n - a \equiv -a$, so $(n - a)^2 \equiv (-a)^2 \equiv a^2$. □

We know two square roots of unity $\pmod{n}$, viz ± 1 . Now consider these:

Example: Find the square roots of 4 $\pmod{7}$.

Answer: The square roots of 4 (mod 7) are 2, 5 (two square roots). Note that $5 \equiv -2 \pmod{7}$

Example: Find the square roots of 6 (mod 12).

Answer: 6 has no square roots (mod 12).

Example: Find the square roots of 1 (mod 12)

Answer: The square roots of 1 (mod 12) are 1, 5, 7, 11. (Unlike the situation with real or complex numbers, we've found a case where a number has 4 square roots!).

Exercise 3: Find the square roots of 9 modulo 36

I found the answers to these examples by computing the squares of the integers $0-3 \pmod{7}$ and $0-6 \pmod{12}$ and using the theorem above. This doesn't seem to be a very sensible method of finding square roots (mod n) for large n . Two obvious questions arise : When can we find square roots of $x \pmod{n}$? How many square roots does x have (mod n)? These turn out to be non-trivial questions, which we won't address. For moderate values of n , brute strength calculation will be adequate for our purposes.

3.2 Quadratic equations mod p

To solve the ordinary quadratic equation $x^2 - 5x + 6 = 0$ by factorisation, we could write $x^2 - 5x + 6 = 0 \equiv (x-2)(x-3) = 0 \implies x-2 = 0$ or $x-3 = 0$ thus $x = 2, 3$.

This technique uses an important consequence of the field properties: if $ab = 0$, then $a = 0$ or $b = 0$.

Theorem 3.2. $ab \equiv 0 \pmod{p}$ implies that $a \equiv 0$ or $b \equiv 0$.

Proof. Suppose $ab \equiv 0$. Either $a = 0$, in which case we have nothing more to prove, or $a \not\equiv 0$. If $a \not\equiv 0$, then a^{-1} exists because the integers (mod p) form a field. In this case, $ab \equiv 0 \implies a^{-1} \cdot a \cdot b \equiv a^{-1} \cdot 0 \implies b \equiv 0$. $\square$

So we can solve quadratic equations (mod p) by factorisation just as we would with real numbers. And since the derivation of the formula for the roots of a quadratic equation depends only on the field properties, we can also solve quadratic equations (mod p) by the quadratic formula-provided $p \neq 2$, in which case the derivation breaks down. (Why does it break down?)

Example: Solve $x^2 + x - 2 \equiv 0 \pmod{5}$

Solution: Method 1: $x^2 + x - 2 = (x-1)(x+2)$, so $x \equiv 1, -2$, ie, $x = 1$ or 3 .

Method 2: By the quadratic formula,

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-1 \pm \sqrt{1^2 - 4 \cdot 1 \cdot (-2)}}{2} \\ &= \frac{-1 \pm \sqrt{1+8}}{2} \equiv \frac{-1 \pm \sqrt{4}}{2} \pmod{5}. \end{aligned}$$

Note that $2^2 \equiv 3^2 \equiv 4 \pmod{5}$ and that $2^{-1} \equiv 3 \pmod{5}$, so that

$$x \equiv \frac{-1+2}{2} \text{ or } \frac{-1+3}{2} \equiv 3 \cdot 1 \text{ or } 3 \cdot 2 \equiv 3 \text{ or } 1.$$

Method 3: Tabulate all possible values of $x^2 + x - 2$ for various values of x :

x	x^2	x	2	
0	0	+	0	- 2 $\neq$ 0
1	1	+	1	- 2 = 0
2	4	+	2	- 2 $\neq$ 0
3	4	+	3	- 2 = 0
4	1	+	4	- 2 $\neq$ 0

So the solution is $x = 1$ or $x = 3$ as before.

Exercise 4: Solve $x^2 + 2x + 2 \equiv 0 \pmod{5}$.

Can you factorise this? Does it help to notice that $+2 \equiv -3 \pmod{5}$? Solve at least by exhaustion and by the quadratic formula, and compare your answers.)

Exercise 5: Solve $x^2 - 4x - 5 \equiv 0 \pmod{11}$

3.3 Quadratic equations mod n (n composite)

If n is composite, then we can use neither factorisation nor the quadratic formula to solve quadratic equations mod n (why not?); but we can solve such congruences by tabulation.

Example: Solve $x^2 + x \equiv 0 \pmod{6}$ Solution: Tabulating all possible values, we get:

so the solutions are $x = 0, 2, 3, 5$.

Notice that the situation is very different from what happens with real or complex numbers. In those cases, a quadratic equation could have no more than 2 solutions. In this simple example we have a quadratic equation with 4 solutions.

Exercise 6: How many solutions are there to the equation $x^2 + x \equiv 0 \pmod{30}$?

Exercise 7: Solve $x^2 + x + 6 \equiv 0 \pmod{12}$.

x	x^2	$x^2 + x$
0	0	0
1	1	2
2	4	0
3	3	0
4	4	2
5	1	0

Exercise 8: Construct some quadratic congruences that you expect to have exactly two solutions. Solve them.

Exercise 9: Construct some quadratic congruences that might not have exactly two solutions. Solve them.

4 Tests for divisibility

Notation: We could develop the following section formally for an integer made up of an arbitrary number of digits (using sigma notation), but for simplicity we'll write, e.g., $abcde$ for the number whose digits are a, b, c, d, e , i.e. the number $10^4 \cdot a + 10^3 \cdot b + 10^2 \cdot c + 10 \cdot d + e$.

Remember that a number x is divisible by n if and only if $x \equiv 0 \pmod{n}$. Explain why!

Divisibility by 3: Since $10 \equiv 1 \pmod{3}$, $10^n \equiv 1 \pmod{3}$ for any positive integer n . So

$$abcde = 10^4 \cdot a + 10^3 \cdot b + 10^2 \cdot c + 10 \cdot d + e \equiv a + b + c + d + e.$$

Therefore $abcde \equiv 0 \pmod{3}$ and so is divisible by 3 if and only if the sum of its digits $(a + b + c + d + e) \equiv 0 \pmod{3}$, i.e., is divisible by 3.

Divisibility by 9: It can similarly be shown that a number is divisible by 9 if and only the sum of its digits is divisible by 9.

Divisibility by 4: Since $100 \equiv 0 \pmod{4}$,

$$\begin{aligned} abcde &= 10^4 \cdot a + 10^3 \cdot b + 10^2 \cdot c + 10 \cdot d + e \\ &= 100(102a + 10b + c) + 10d + e \equiv 10d + e \pmod{4}, \end{aligned}$$

so the integer $abcde$ is divisible by 4 if and only if its last two digits (ie the two digit number $10d + e$) are divisible by 4. e.g., since 28 is divisible by 4, 12345628 is divisible by 4 (can you see that 12345678 is not divisible by 4?)

Divisibility by 8: It can similarly be shown that a number is divisible by 8 if and only if the number formed by its last three digits is divisible by 8.

Divisibility by 11: Note that $10 \equiv -1 \pmod{11}$, so $100 \equiv 1 \pmod{11}$. Hence for all natural numbers n , $10^{2n} \equiv 1$, and $10^{2n+1} \equiv -1$. So

$$abcde = 10^4 \cdot a + 10^3 \cdot b + 10^2 \cdot c + 10 \cdot d + e \equiv a - b + c - d + e \pmod{11},$$

so $abcde$ is divisible by 11 if and only if the sum of the digits in the even positions minus the sum of the digits in the odd positions is divisible by 11. e.g., 19283748 is divisible by 11 because $(1 + 2 + 3 + 4) - (9 + 8 + 7 + 8) = 10 - 32 = -22$, which is divisible by 11.

Exercise 10: Use congruences to prove that all perfect squares end in 0,1,4,5,6, or 9.

Exercise 11: Prove that all perfect squares are congruent to 0 or 1 (mod 4).

Exercise 12: Prove that all primes are congruent to 1 (mod 6) or to 5 (mod 6).

Exercise 13: Consider the alternative argument:

$$19283748 \longrightarrow 19 + 28 + 37 + 48 = 132 \longrightarrow 01 + 32 = 33,$$

which is divisible by 11, hence 19283748 is divisible by 11. Use congruences to determine whether this is a valid divisibility test or not.

5 More on Finding Inverses (mod p)

Discovering an efficient general method of finding inverses (mod p) requires knowledge of further properties of ordinary integers. We will first consider "Euclid's Algorithm", which is important in its own right, and will then outline, how Euclid's Algorithm can be used in finding $x^{-1} \pmod{p}$.

5.1 Greatest Common Divisor (gcd)

Notation: For natural numbers a and b , the notation $a|b$ (a divides b) means that a is a divisor of b , or that b is a multiple of a . $a|b$ if and only if there is a natural number k such that $ka = b$.

For example, $3|21$ because $7 \cdot 3 = 21$.

The gcd (greatest common divisor, sometimes known as the hcf, or highest common factor) of two natural numbers a and b is the natural number g with the property that $g|a$, $g|b$, and any other natural number h such that $h|a$ and $h|b$ also has the property that $h|g$.

Intuitively, the gcd of two natural numbers is the "largest" natural numbers which divides both of them. We write (a, b) for the gcd of a and b .

The definition of gcd doesn't prove that it exists for every pair of natural numbers a, b . We'll see soon that it always exists.

5.2 Finding gcd by factorisation

If a and b are reasonably small, it is easy to find (a, b) by reducing a and b to their prime factors and checking which factors are common to both.

Example Find the gcd of 220 and 840.

Solution: $220 = 2 \cdot 110 = 2 \cdot 2 \cdot 55 = 2 \cdot 2 \cdot 5 \cdot 11$, and we can similarly discover that $840 = 2 \cdot 2 \cdot 2 \cdot 3 \cdot 5 \cdot 7$. Hence, $(220, 840) = 2 \cdot 2 \cdot 5 = 20$.

This technique is easy to apply if the numbers are small (and have conveniently small prime factors), but can rapidly become infeasible. Could you readily find, for example, the gcd of 31683 and 34547?

5.3 The "division algorithm"

The "division algorithm" tells us that for any integers $a > b > 0$, we can find integers q and r such that $a = bq + r$, with $q > 0$ and $0 \leq r < b$.

For example, given $a = 17$, $b = 5$, we can write $17 = 5 \cdot 3 + 2$. Note that 15 ($= 5 \cdot 3$) is the largest multiple of 5 which is not greater than 17, and 2 is the remainder when we divide between 17 by 5.

5.4 Euclid's algorithm

If an integer d divides any two of the integers a , b , c , and $a + b = c$, then d divides the third of them also. This property, together with the division algorithm, is the basis for a technique (known as Euclid's Algorithm) for finding the gcd of any two positive integers. (With the aid of even the cheapest pocket calculator, finding the gcd of a pair of, say, eight digit numbers in this way is not too painful, but we'll restrict ourselves to smaller numbers to illustrate the important principles.)

Preliminary Example: Find the gcd of 220 and 840.

We know the answer from above, now we'll do it "properly":

$$840 = 3 \cdot 220 + 180 \quad 220 = 1 \cdot 180 + 40 \quad 180 = 4 \cdot 40 + 20 \quad 40 = 2 \cdot 20$$

We find that $\text{gcd} = 20$ as before-but this time the method generalises:

Example: Find the gcd of 546 and 4620.

Solution: Suppose that g is the required gcd. Dividing 4620 by 546, we find that $4620 = 546 \cdot 8 + 252$. Now since $g \mid 4620$ and $g \mid 546$, it follows that $g \mid 252$. So g is a common divisor of 546 and 252.

Repeat the process. Dividing 546 by 252, we find that $546 = 2 \cdot 252 + 42$. Once again, since $g \mid 546$ and $g \mid 252$, $g \mid 42$. So g is a common divisor of 252 and 42.

Now carry out the division algorithm again. $252 = 42 \cdot 6$, so 42 is a common divisor of 42 and 252. Since we have no remainder at this stage, we conclude that $g = 42$ is a common divisor of 546 and 4620.

But how do we know that 42 is the greatest common divisor? Suppose some other common factor h divides both 4620 and 546. Then from the first paragraph of our solution, $h|252$. From the second paragraph, since $h|546$ and $h|252$, then $h|42$. So any other divisor of both 4620 and 546 also divides 42, so 42 is the gcd.

All of this writing is tedious, so once we understand the idea we can streamline the process using a standard layout. Notice that to find the gcd our only real interest at each stage is the remainder. (The quotient at each stage also provides us with useful information as shown on page ?? of these notes.) We can keep track of each quotient and remainder by arranging the above calculation thus:

$$\begin{array}{r}
 546) 4620 \ (8 \\
 \underline{4368} \\
 252) 546 \ (2 \\
 \underline{504} \\
 42) 252 \ (6 \\
 \underline{252} \\
 0
 \end{array}$$

and now that we have a 0 remainder, we can read off the gcd.

Without a calculator, each division in this table can be expanded to show all the steps in the "long division" process we learnt at school, but those steps play no part in our use of the results.

Exercise 14: Use Euclid's Algorithm to prove that the gcd of 127 and 179 is 1.

Exercise 15: Find the gcd of 31683 and 34547.

5.5 Outline of the Proof of Euclid's Algorithm - Optional

A formal proof that Euclid's Algorithm works can be produced by replacing 546 and 4620 in the example above by arbitrary numbers a and b , the successive quotients by $q_1, q_2, \dots$, and the successive remainders by $r_1, r_2, \dots$. The argument regarding common divisors is the same as in that example. The remaining step in the proof is to establish that the process terminates; this follows from the fact that each remainder becomes the divisor at the next step, and that these remainders steadily decrease, ie, that $0 = r_{i+1} < r_i$. But the values of r cannot become negative, so process must stop when the remainder becomes 0. This proof would commence thus:

Suppose that a, b are given integers with $a > b > 0$, and we wish to find the gcd g . Then by the

division algorithm we can write:

$$\begin{array}{lll}
 a = q_1b + r_1 & \text{with } 0 \leq r_1 < b & \text{and if } g|a \text{ and } g|b, \text{ then } g|r_1 \\
 b = q_2r_1 + r_2 & \text{with } 0 \leq r_2 < r_1 & \text{and if } g|b \text{ and } g|r_1, \text{ then } g|r_2 \\
 r_1 = q_3r_2 + r_3 & \text{with } 0 \leq r_3 < r_2 & \text{and if } g|r_1 \text{ and } g|r_2, \text{ then } g|r_3 \\
 \vdots & \vdots & \vdots \\
 r_{n-1} = q_{n+1}r_n & & \text{remainder} = 0, \text{ and } r_n \text{ is the gcd of } a \text{ and } b.
 \end{array}$$

If $g = (a, b)$, then there exist integers x, y such that $xa + yb = g$.

Theorem 5.1. *If $g = (a, b)$, then there exist integers x, y such that $xa + yb = g$.*

We won't prove this theorem formally. Essentially, x and y can be found by "working backwards" through Euclid's Algorithm; the following calculation illustrates the technique:

In the example above we have,

$$\begin{array}{lll}
 4620 = 8 \cdot 546 + 252 & \text{from which we may write} & 252 = 4620 - 8 \cdot 546 \\
 546 = 2 \cdot 252 + 42 & & 42 = 546 - 2 \cdot 252 \\
 252 = 6 \cdot 42. & &
 \end{array}$$

So,

$$42 = 546 - 2 \cdot (4620 - 8 \cdot 546) = 546 - 2 \cdot 4620 + 16 \cdot 546 = 17 \cdot 546 - 2 \cdot 4620.$$

Exercise 16: Find integers x, y such that $x \cdot 127 + y \cdot 179 = 1$.

5.6 Using the gcd to find $a^{-1} \pmod{p}$

If p is prime and $a < p$, then $\gcd(a, p) = 1$ (why?). So by the theorem above there exist (and we can find) integers x, y such that $xa + yp = 1$. Now take congruences mod p in this equation: Then $xa + 0 \equiv 1 \pmod{p}$, or $xa \equiv 1 \pmod{p}$, and so $x = a^{-1} \pmod{p}$.

Example: Find integers x, y such that $x \cdot 127 + y \cdot 179 = 1$. Hence deduce the value of $127^{-1} \pmod{179}$.

Solution: From **Exercise 16**, or using elementary arithmetic to verify the following, we see that $22 \cdot 179 - 31 \cdot 127 = 1$. Taking congruences $\pmod{179}$, this gives $0 - 31 \cdot 127 \equiv 1 \pmod{179}$, or $(-31) \cdot 127 \equiv 1 \pmod{179}$, so $127^{-1} \pmod{179} \equiv -31 \equiv 148$.

6 Additional Exercises

Exercise 17: (1994 Mid-Year Exam) Find the square roots of $4 \pmod{15}$.

Exercise 18: Prove that $28^{-1} \pmod{37} = 4$. Hence solve the congruence $28x \equiv 19 \pmod{37}$.

Exercise 19: (1994 Mid-Year Exam) Find an integer n such that $2^n \equiv 1 \pmod{31}$. Deduce also the remainder when 2^{100} is divided by 31.

Exercise 20: (1992 Mid-Year Exam) Suppose that $0 \leq y \leq 999$ and $143y$ is congruent to 41 modulo 1000. What is y ?

Exercise 21: (1994 Mid-Year Exam) Solve the congruence $x^2 + x - 2 \equiv 0 \pmod{5}$.

Exercise 22: (1990 Mid-Year Exam) Solve $3x^2 + 5x + 1 = 0$ in $\mathbb{Z}_{13}$.

Exercise 23: (1994 Mid-Year Exam) Use Euclid's Algorithm to show that the greatest common divisor (gcd) of 20 and 32 is 4.

Exercise 24: (1994 Mid-Year Exam) Consider the five-digit number $n = abcde$ in decimal notation.

- (i) Give a proper expression of n in terms of a, b, c, d, e .
- (ii) Show that $n - (a + b + c + d + e)$ is divisible by 3.
- (iii) Formulate a suitable rule for divisibility of five-digit numbers by 3.
- (iv) Generalise your divisibility rule to numbers with any number of digits.

Exercise 25: Given that $4 \cdot 79 - 9 \cdot 35 = 1$, find the inverse of 35 $\pmod{79}$ and hence solve the congruence $35x \equiv 3 \pmod{79}$.

Exercise 26: (1994 End-of-Year Exam) Use congruences mod 4 to prove that if $3^x + 4^y = 5^z$ for some positive integers x, y , and z , then x is even.

Exercise 27: (1994 End-of-Year Exam) Explain briefly why the following "solution" is unsatisfactory:

Problem: Solve the congruence $x^2 \equiv 3x - 2 \pmod{6}$.

"Solution": Working mod 6, if $x^2 \equiv 3x - 2$, then $x^2 - 3x + 2 \equiv 0$, so $(x - 1)(x - 2) \equiv 0 \pmod{6}$, and the solution is $x \equiv 1$ or $2 \pmod{6}$.

Exercise 28: (1995 Mid-Year Exam) Prove that if n is odd, $n^2 \equiv 1 \pmod{8}$.

Exercise 29: (1995 Mid-Year Exam) A number N has the form $N = 2^m + 1$. Use congruences $\pmod{3}$ to prove that if m is odd, then $3 \mid N$.

7 Exercises: Selected Hints, Answers, or Outline Solutions

Exercise 1: $2 \cdot 7 = 14 \equiv 1 \pmod{13}$, so 2 and 7 are inverses of each other. $3 \cdot 9 = 27 \equiv 1 \pmod{13}$, so 3 and 9 are inverses of each other. Similarly we get 5 and 8, 4 and 10, and 6 and 11. These 5 pairs account for 10 integers. $1^{-1} = 1$, $(-1)^{-1} = -1$, and 0^{-1} doesn't exist.

Exercise 2: One solution: If $6x \equiv 4 \pmod{13}$, $11 \cdot 6x \equiv 11 \cdot 4 \pmod{13}$, so $x \equiv 5 \pmod{13}$.

Exercise 3: It might appear that we need to test 36 numbers. But since $(n-a)^2 \equiv a^2$, we can halve this number, and since 9 is odd we need test only odd numbers as possible square roots. So we need test only 1, 3, 5, 7, 9, 11, 13, 15, 17, and we find that 9 has six square roots $\pmod{36}$: 3, 9, 15, 21, 27, 33. Did you notice that all of these square roots have the form $6k+3$, i.e., they are all congruent to 3 $\pmod{6}$? Why is this so?

Exercise 4: $x^2 + 2x + 2 \equiv x^2 + 2x - 3$, so $(x+3)(x-1) \equiv 0$, $\implies x \equiv 1, -3$, or $x \equiv 1, 2 \pmod{5}$

Exercise 5: The solution is $x \equiv 5, 10$.

Exercise 6: You can answer this question by tabulating all values of $x^2 + x \pmod{30}$. Alternatively, we want all solutions to $x(x+1) \equiv 0 \pmod{30}$, so we need all values of x for which $x(x+1)$ is a multiple of 30. I noticed that 30 has prime factors of $2 \cdot 3 \cdot 5$, so I wanted $x(x+1)$ to have all these prime factors. I thus guessed that possible values of x may be 0, 5, 9, 14, 15, 20, 24, 29, and found that these values worked. Note that using this approach it isn't easy to see whether we've found all the solutions, and note too that since 30 is composite we can't use either the quadratic formula or factorisation.

Exercise 7: Tabulate the values of $x^2 + x + 6$. The solution is $x = 2, 5, 6, 9$.

Exercise 8: Try using congruences $\pmod{p}$.

Further exercise: Construct, if possible, quadratic congruences with prime moduli which have (i) 0 solutions; (ii) exactly one solution; (iii) more than two solutions. It may help to remember that if p is prime, quadratic congruences $\pmod{p}$ can be solved by the quadratic formula-but do all numbers $\pmod{p}$ have square roots? (You can check your answers to this further exercise as usual by solving each quadratic using at least two different methods).

Exercise 9: Try using congruences $\pmod{n}$, where n is composite.

Exercise 10: Use the fact that "end in ..." is equivalent to " $\equiv \dots \pmod{10}$ ".

Exercise 11: Every integer is expressible either as $2n$ or $2n+1$. $(2n)^2 = 4n^2 \equiv 0 \pmod{4}$, and $(2n+1)^2 = 4n^2 + 4n + 1 \equiv 1 \pmod{4}$.

Can you improve on this result for odd squares?

Exercise 12: All integers congruent to 0 $\pmod{6}$ are divisible by 6; all integers congruent to 2 or 4 $\pmod{6}$ are divisible by 2; and all integers congruent to 3 $\pmod{6}$ are divisible by 3.

Exercise 13: $19283748 = 19 \cdot 10^6 + 28 \cdot 10^4 + 37 \cdot 10^2 + 48 \equiv 19 + 28 + 37 + 48 \pmod{11}$ since 10^2 (and hence 10^{2n}) are congruent to 1 $\pmod{11}$. Now repeat the argument on 0132. The test is a valid alternative.

Exercise 14: $179 = 1 \cdot 127 + 52$; $127 = 2 \cdot 52 + 23$; $52 = 2 \cdot 23 + 6$; $23 = 3 \cdot 6 + 5$;
 $6 = 1 \cdot 5 + 1$; $5 = 5 \cdot 1$.

Exercise 15: The gcd of 31683 and 34547 is 179: $34547 = 1 \cdot 31683 + 2864$; $31683 = 11 \cdot 2864 + 179$; $2864 = 16 \cdot 179$

Exercise 16: From Ex 14, $179 - 1 \cdot 127 = 52$; $127 - 2 \cdot 52 = 23$; $52 - 2 \cdot 23 = 6$; $23 - 3 \cdot 6 = 5$;
 $6 - 1 \cdot 5 = 1$. Now substituting from the bottom up, $1 = 6 - 1 \cdot 5 = 6 - (23 - 3 \cdot 6) = 4 \cdot 6 - 1 \cdot 23 =$
 $4(52 - 2 \cdot 23) - 1 \cdot 23 = 4 \cdot 52 - 9 \cdot 23 = 4 \cdot 52 - 9(127 - 2 \cdot 52) = 22 \cdot 52 - 127 = 22(179 - 1 \cdot 127) -$
 $9 \cdot 127 = 22 \cdot 179 - 31 \cdot 127$.

Exercise 17: Testing the numbers 1 - 7 by trial and error is sufficient to show that the square roots of 4 $\pmod{15}$ are 2, 7, i.e., 2, 7, 8, 13.

Exercise 18: Use $28 \cdot 4 = 112 = 3 \cdot 37 + 1$. So modulo 37, $28x \equiv 19 \implies 4 \cdot 28x \equiv 4 \cdot 19 \implies x \equiv 2$.

Note that the solution " $28 \cdot 2 = 56 \equiv 19 \pmod{37}$ ", so $x = 2$ " solves the congruence, but the question said "Hence solve ...". The object of this exercise was to use the inverse of 28 to solve the congruence; the fact that it can be solved another way provides a useful check on the solution, but it doesn't answer the question.

Exercise 19: You should be familiar with the small powers of 2; if not, trial and error soon gives $n = 5$ as a suitable value. Then $2^{100} = (2^5)^{20} \equiv 1 \pmod{31}$, whence the remainder is 1.

Exercise 20: I remember that $7 \cdot 14 = 98$, and so I see that $7 \cdot 140 = 980$. How does this help me to find the inverse of 143 $\pmod{1000}$?

Exercise 21: This exercise is most easily done by factorisation: $x^2 + x - 2 = (x + 1)(x - 2)$, etc. Alternatively, since the modulus 5 is prime we may use the quadratic formula, or we can tabulate $x^2 + x - 2$ for all values of x . Do it all three ways and reconcile your answers. NB: in 1994 a very high proportion of students who used tabulation made careless errors-transcription errors, trivial arithmetic errors (like $12 + 1 = 0$), etc. Those who solved by factorisation had most success, provided they remembered that they were solving a congruence equation and not an ordinary quadratic equation.

Exercise 22: Routine. The modulus might be large enough to encourage you to use the quadratic formula. Try it at least two ways and compare your answers. No answer is given-verify your own solutions, as you must always do.

Exercise 23: The gcd is obviously 4, so there is no credit for finding the number 4. The point of the question is for you to demonstrate that you understand Euclid's Algorithm; if you use any shorthand method of setting out, you must make it quite clear that you could use the algorithm to find the gcd of any pair of integers (cf **Exercise 15 above**). A useful check is to ask yourself whether your

presentation like yours would provide a suitable explanation of the solution of **Exercise 15** in these notes.

Exercise 24: In 1994 this question was very badly done in two respects: many students made no attempt to explain their working (how can you "show that ..." without explanation?); and many students who trusted their memories misquoted a rule for divisibility so badly that one or two trivial examples would have readily shown that they were wrong. Here is an outline of a possible solution: (i) $n = 104a + 103b + 102c + 10d + e$; (ii) $n = a + b + c + d + e + 9999a + 999b + 99c + 9d$, so $n \equiv a + b + c + d + e \pmod{3}$, so $n - (a + b + c + d + e) \equiv 0 \pmod{3}$, i.e., $n - (a + b + c + d + e)$ is divisible by 3; (iii) since $n \equiv (a + b + c + d + e) \pmod{3}$, any five digit number n is divisible by 3 if and only if the sum of its digits is divisible by 3; (iv) any integer is divisible by 3 if and only if the sum of its digits is divisible by 3 (the proof above is easily modified to apply to number with more or fewer digits than 5).

Exercise 25: Modulo 79, $4 \cdot 79 - 9 \cdot 35 = 1$ becomes $4 \cdot 0 - 9 \cdot 35 \equiv 1$, or $(-9) \cdot 35 \equiv 1$. So $35^{-1} \equiv -9 \equiv 70 \pmod{79}$. Hence $35x \equiv 3 \implies 70 \cdot 35x \equiv 70 \cdot 3$, or $x \equiv 210 \equiv 52 \pmod{79}$.

Exercise 26: The first and most important comment of all is "read the question!" The question says "prove that"; a proof requires English sentences, not a meaningless mass of symbols. Next, the question says "Use congruences $\pmod{4}$...", so any sensible solution will start by taking congruences $\pmod{4}$: Now if $3^x + 4^y = 5^z$, then $3^x \pmod{4} + 4^y \pmod{4} \equiv 5^z \pmod{4}$, so $(-1)^x + 0 \equiv 1^z$ i.e., $(-1)^x \equiv 1 \pmod{4}$, and this holds if and only if x is even. If you are want to go into even more detail in this final step, note that if x is odd, then $x = 2n + 1$, say, and so $(-1)^x = (-1)^{2n+1} = (-1)^{2n} \cdot (-1) = -1 \equiv 3 \pmod{4}$. However, my "if and only if" statement above is sufficient.

Exercise 27: Once again, the key comment is "read the question!" This question doesn't ask for a correct solution to the "Problem", it asks what is wrong with the given "Solution". Once again, the way to explain what is wrong with something is by writing sentences, not meaningless symbols. In particular, showing that $x \equiv 4$ and $x \equiv 5$ are also solutions to the given equation says little more about why the given "solution" is unsatisfactory than a declaration that "it doesn't work", and it certainly doesn't explain why the "solution" would have been unsatisfactory even if it gave correct roots to the equation. The point of the question was to allow students to demonstrate that they know that integers modulo 6 do not form a field (because 6 is composite), and that therefore solution of quadratics by factorisation is invalid because not all integers mod 6 have inverses; this means that some solutions may be lost (as indeed they are in this case).

Exercise 28: (First of all, note that there is an infinite number of odd integers, so it is impossible to answer this question by referring to particular examples.) Every odd integer m is congruent to 1 $\pmod{2}$, and so can be written in the form $n = 2m + 1$. Then $n^2 = 4m^2 + 4m + 1 = 4m(m + 1) + 1$. Now one of m , $m + 1$ is even, so $m(m + 1)$ is even, say $m(m + 1) = 2r$. So $n^2 = 4 \cdot m(m + 1) + 1 = 4 \cdot 2r + 1 = 8r + 1 \equiv 1 \pmod{8}$.

Exercise 29: As in **Exercise 28**, there is an infinite number of such numbers N , and particular cases prove nothing. If m is odd, say $m = 2n + 1$, then $N = 2^{2n+1} + 1 = 2 \cdot 2^{2n} + 1 = 2 \cdot 4^n + 1 \equiv 2 \cdot 1^n + 1 \pmod{3} \equiv 2 + 1 \pmod{3} \equiv 0 \pmod{3}$, so N is divisible by 3 as required.